

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Melting point of Boron – 2453 K, Iron – 1808 K, Silicon – 1693 K and Aluminium – 933 K. Hence, among the given options, Boron has highest melting point.

Alloys

Notes

- An alloy is a uniform mixture.
- It is made up of two or more chemical elements, of which at least one is a metal.
- An alloy has properties different from the metal it is made of.
- Steel is an alloy which is made by mixing carbon (0.5% to 1.5%) with iron.
- Sterling silver is an alloy of silver that consists of 92.5% pure silver and 7.5% of other metal usually copper. Sterling silver is used in jewellery, silverware and decorations. Alloying with copper maintains the silvery color of metal while increasing stability and rigidity.

List of some alloys, their components and uses

S.N.	Name of Alloy	Components	Uses
1.	Brass	Copper + Zinc (Cu) + (Zn)	In making utensils and decorative materials.
2.	Steel	Iron + Carbon (Fe) + (C)	In buildings, infrastructure, tools, ships, automobiles, appliances and weapons.
3.	Stainless Steel	Steel + Chromium + Nickel + Carbon etc.	In making utensils, surgical tools and medical equipments, in architect, automotive and aerospace applications etc.
4.	Bronze	Copper + Tin (Cu) + (Sn)	In making bronze bearings, spark free tools, coins, medals, sculptures and musical instruments.

5.	Solder	Lead + Tin (Pb) + (Sn)	Soldering is a process of joining two or more metals by heating.
6.	Duralumin (a hard, light alloy)	Aluminium + Copper + Magnesium + Manganese	Widely used in aircraft construction and in making cookwares.
7.	German Silver	Copper + Nickel + Zinc	It is extensively used because of its hardness, toughness and resistance to corrosion for articles such as tableware, marine fittings and plumbing fixtures. Because of its high electrical resistance it is also used in heating coils.
8.	Alnico	Iron + Aluminium + Nickel + Cobalt	Used to make permanent magnet.
9.	Nichrome	Nickel + Iron + Chromium + Manganese	Used in heating elements.

Karat/Carat

- A karat, when used with gold, is a unit of purity.
- 24-karat gold is pure (100%) gold.
- Usually to make jewellery some metals like copper or silver is mixed with it, because pure gold is too soft.
- 18-karat gold is 75% gold and 25% other metals, and so on.
- 1 gram of 12-karat gold contains 50% or 0.5 grams of pure gold.
- Each karat indicates 1/24th of the whole.
- While 'karat' is commonly used in the USA to describe gold purity, 'carat' is also used in some regions (like, the UK, Australia and India) to describe gold purity.
- The standard unit of weight for gemstones, including diamonds, is the carat.
- One carat is defined as exactly 200 milligrams or 0.2 grams.

Question Bank

1. Out of the following which is not an alloy?

- (a) Steel (b) Brass
(c) Bronze (d) Copper

M.P.P.C.S. (Pre) 1998

Ans. (d)

Copper is a metal, however steel, brass and bronze are alloys.

2. Alloy of which metal is used to make aeroplane and parts of the compartment of the train?

- (a) Copper (b) Iron
(c) Aluminium (d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Low density and strength make aluminium ideal for construction of aircraft, lightweight vehicles and ladders. An alloy of aluminium called Duralumin is often used instead of pure aluminium because of its improved properties. Easy shaping and corrosion resistance make aluminium a good material for drinking cans and roofing materials.

3. Mercury-metal mixture is –

- (a) High colour alloy
(b) Carbon mixed alloy
(c) Mercury mixed alloy
(d) Highly resisting alloy for malfriction

39th B.P.S.C. (Pre) 1994

Ans. (c)

An amalgam is a substance formed by the reaction of mercury with another metal. Almost all metals can form amalgams with mercury, the notable exception being iron.

4. Which one of the following metals does not form amalgam ?

- (a) Zinc (b) Copper
(c) Magnesium (d) Iron

I.A.S. (Pre) 1998

Ans. (d)

See the explanation of above question.

5. Which one of the following alloys is called an amalgam ?

- (a) Zinc – Copper
(b) Copper – Tin
(c) Mercury – Zinc
(d) Lead – Zinc

U.P.P.C.S. (Mains) 2012

Ans. (c)

An amalgam is a substance formed by the reaction of Mercury with another metal.

6. Solder used in soldering metal pieces is an alloy of :

- (a) Tin and Lead (b) Tin and Copper
(c) Tin, Zinc and Copper (d) Tin, Lead and Zinc

U.P.P.C.S. (Pre) 2005

U.P.P.C.S. (Pre) 1996

Ans. (a)

The common composition of solder is around 32% Tin and 68% Lead. This combination has a low melting point and is useful for soldering components that are sensitive to heat.

7. Bronze is an alloy of –

- (a) Copper and Tin (b) Copper and Silver
(c) Copper and Zinc (d) Copper and Lead

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

Bronze is an alloy made up of Copper and another metal Tin. Compositions may vary but most modern bronze is around 88% Copper and 12% Tin.

8. Bronze is an alloy of copper and –

- (a) Tin (b) Aluminium
(c) Silver (d) Nickel

U.P. P.C.S. (Mains) 2016

Ans. (a)

See the explanation of above question.

9. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I	List-II
A. German silver	1. Tin
B. Solder	2. Nickel
C. Bleaching powder	3. Sodium
D. Hypo	4. Chlorine

Code :

	A	B	C	D
(a)	1	2	4	3
(b)	2	1	3	4
(c)	1	2	3	4
(d)	2	1	4	3

I.A.S. (Pre) 2000

Ans. (d)

German Silver or Nickel silver is an alloy consisting of Copper (around 60%), Zinc (around 20%) and Nickel (around 20%). It doesn't have Silver. Solder of Tin and Lead is known as soft solder. The chemical name of bleaching powder is calcium hypochlorite $\text{Ca}(\text{ClO})_2$. Sodium Thiosulphate (Hypo) is a crystalline white smellless solid.

10. Percentage of Silver in German Silver is :

- (a) 1% (b) 5%

- (c) 0% (d) None of these

U.P.P.C.S. (Mains) 2003

Ans. (c)

See the explanation of above question.

11. How much percentage of silver is found in German Silver?

- (a) 20% (b) 30%
(c) 60% (d) None of these

Chhattisgarh P.C.S. (Pre) 2020

Ans. (d)

See the explanation of above question.

12. Which one of the following elements is not present in the German Silver?

- (a) Copper (b) Aluminium
(c) Zinc (d) Nickel

R.A.S./R.T.S.(Pre) 2007

Ans. (b)

See the explanation of above question.

13. Which of the following alloys has a maximum percentage of Copper ?

- (a) Brass (b) Bronze
(c) German Silver (d) Gun Metal

U.P. Lower Sub. (Spl.) (Pre) 2008

Ans. (b)

Brass consists about 68-71% Copper and rest is Zinc. Bronze consists about 88% Copper and 12% Tin. German silver has almost 50% Copper. Gunmetal consists about 85% Copper, 5% Tin, 5% Lead, 5% Zinc.

14. Which metal exists as a common component in Brass, Bronze and German Silver?

- (a) Antimony (b) Copper
(c) Tin (d) Zinc

U.P. Lower Sub. (Pre) 2002

Ans. (b)

See the explanation of above question.

15. Brass is an alloy of –

- (a) Copper and Nickel (b) Nickel and Zinc
(c) Copper and Zinc (d) Iron and Nickel

47th B.P.S.C. (Pre) 2005

R.A.S./R.T.S. (Pre) 1997

Ans. (c)

See the explanation of above question.

16. Which among the following are constituents of brass?

- (a) Zinc and copper
(b) Iron and zinc
(c) Copper and nickel
(d) Iron and copper
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

See the explanation of above question.

17. Brass is a combination of what metals?

- (a) Copper and Nickel
(b) Copper and Tin
(c) Copper and Zinc
(d) Copper + Zinc and Nickel

Chhattisgarh P.C.S. (Pre) 2011

Uttarakhand P.C.S. (Pre) 2002

U.P.P.C.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

18. Which pair of following is of Copper alloys?

- (a) Brass and Invar
(b) Bronze and Nichrome
(c) Brass and Bronze
(d) None of the above

Uttarakhand P.C.S. (Pre) 2016

Ans. (c)

See the explanation of above question.

19. Check these statements :

- (i) Brass is an alloy of Copper and Zinc.
(ii) Magnetite is the main ore of Aluminium.
(iii) Amalgams are that substances which mainly involve mercury.
(iv) Potassium Nitrate (KNO_3) is a chemical used in photography.

Which of the above are true ?

Code :

- (a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (i), (iii) and (iv)

U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (b)

Statement (i) and (iii) are correct but statement (ii) and (iv) are wrong because Magnetite (Fe_3O_4) is an ore of Iron and Silver Bromide (AgBr) is used in photography.

20. Brass gets fade colour in air due to the presence of –

- (a) Oxygen
- (b) Hydrogen Sulphide
- (c) Carbon dioxide
- (d) Nitrogen

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

Hydrogen Sulphide is responsible for fading of Brass.

21. Combination of which substance in the alloy renders stainless steel non-magnetic ?

- (a) Carbon
- (b) Chromium
- (c) Nickel
- (d) Molybdenum

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

Stainless steel becomes non-magnetic by combining the alloy with Nickel, while extra Carbon is mixed to gain more hardness.

22. To make the steel hard requires increase in –

- (a) The quantity of Carbon
- (b) The quantity of Manganese
- (c) The quantity of Silicon
- (d) The quantity of Chromium

42nd B.P.S.C. (Pre) 1997

Ans. (a)

Steels are alloys of Iron and other elements primarily Carbon and are widely used in construction and other area. Carbon mixed in Iron as a hardening agent which strengthens Iron by distorting its crystal lattice. This distortion results in hardening.

23. Steel slag can be the material for which of the following?

1. Construction of base road
2. Improvement of agricultural soil
3. Production of cement

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

Steel slag is an unavoidable by-product in iron and steel production. Steel slags are produced at steel melting shop during steel manufacturing. In steel slags, total iron content varies from around 16% to 25%. It also contains calcium, silicon, manganese, phosphate, etc. It has shown potential for use as a raw mix components up to 10% in the manufacture of cement clinker. Steel slag can also replace granulated blast

furnace slag up to 10% in the manufacture of portland slag cement. Its physical characteristics are similar to air-cooled iron slag and hence it is suitable for use in ready-mixed concrete, concrete products, road bases and similar applications in construction. Steel slag has found use as a barrier material remedy for waste sites where heavy metals tend to leach into the surrounding environment. Steel slag forces the heavy metals to drop out of solution in water runoff because of its high oxide mineral content. It has been used successfully to treat acidic water discharges from abandoned mines. Steel slag can be used to treat acidic soils. According to the Ministry of Steel, the use of steel slag may be one of the most green solutions that may avoid the accumulation of large wastes in steel plants as well as help in the treatment of soil for improving crop productivity at low cost. From the above explanation, it is clear that all of the given statements are correct.

24. Which of the following elements is added to iron to produce steel which can resist high temperature, have high hardness and abrasion resistance ?

- (a) Aluminium
- (b) Chromium
- (c) Nickel
- (d) Tungsten

I.A.S. (Pre) 1996

Ans. (b)

Stainless steel is a steel alloy with Chromium content. The presence of Chromium protects steel from corrosion, rust and make it high- temperature resistant.

25. The important metal used with iron to make it rust free–

- (a) Aluminium
- (b) Carbon
- (c) Chromium
- (d) Tin

Chhattisgarh P.C.S. (Pre) 2011

Ans. (c)

See the explanation of above question.

26. Steel is further processed to obtain stainless steel by adding some element. Which one of the following is not used for this purpose ?

- (a) Carbon
- (b) Silicon
- (c) Manganese
- (d) Nickel

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (b)